

Hsin-Yuan (Samuel) Lee

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Education

University of California, Los Angeles, Los Angeles, CA Sep 2026 – Jun 2028 (Expected)

Master of Science (M.S.) in Computer Science

University of California, San Diego, La Jolla, CA Sep 2022 – Jun 2026

B.S. Data Science; B.S. Cognitive Science (Machine Learning & Neural Computation), **Cum Laude** **GPA: 3.92/4.00**

Technical Skills

Languages: Python, C++, JavaScript, Bash

ML/AI: Deep Learning, Computer Vision, NLP, LLMs, AI Agents, RAG, Transformers, CNNs, Mechanistic Interpretability

Frameworks and Libraries: PyTorch, PyTorch Lightning, scikit-learn, NumPy, pandas, LangChain, FastAPI, Flask

Cloud and Tools: AWS (EC2, S3), Azure OpenAI, Docker, Kubernetes, Jenkins, Git, Gerrit, Unreal Engine 5

Professional Experience

Realtek Semiconductor Corp., Hsinchu, Taiwan Jul 2026 – Sep 2026

Software Engineering Intern

- Built two Jenkins-integrated Bash/Python automation tools (~9,900 lines) replacing a manual Git/Gerrit repo-mirroring workflow, cutting full-repository sync from ~**3 hours** to ~**25 min** across **400+ repositories** used by 30 engineers.
- Wrote SKILL.md specs that let a Claude Code agent drive the workflow; packaged the tools as a versioned skill hub.

Advantech Co., Ltd., Taipei, Taiwan Jun 2023 – Sep 2023

Software Development Intern

- Built a retrieval-augmented generation (RAG) chatbot on Azure OpenAI and Azure AI Search with automated document ingestion, cutting response time from days to under 20 seconds; won the **Best Technical Intern Project** award.
- Designed a punctuation-aware Mandarin NLP tokenizer, lifting ambiguous-query retrieval accuracy from **34% to 76%**.

Research Experience

Gary's Unbelievable Research Unit (Prof. Garrison Cottrell), UC San Diego Aug 2025 – Jun 2026

Undergraduate Research Assistant

- Trained a ResNet-18 with fixation sampling, log-polar transforms, and salience-weighted preprocessing on **76,800 images** of 384 face, dog, and object identities.
- Reproduced the human Face Inversion Effect, matching human accuracy within **6%** in all 4 Yin (1969) conditions.

Qualcomm Institute (Calit2) (Dr. Neil Smith), UC San Diego Aug 2025 – Jun 2026

Research and Development Intern

- Built a real-time 3D avatar with TTS and gesture generation, writing its core in **C++ and Unreal Engine 5**; demoed to **20+ visitors**.
- Designed and deployed a FastAPI and Kubernetes pipeline, reducing avatar response latency by **25%**.

Publications

When AI Explains AI: Understanding and Mitigating the Interpretability Lottery. *ICLR 2027*, under review; first author.

HypoEvolve: Genetic Algorithms Enable Multi-Agent LLMs to Discover Scientific Hypotheses. *ICLR 2027*, under review.

A Mechanistic Explanation for the Inverted Face Effect. *CogSci 2026*, accepted; co-first author.

Projects

Reliability of Sparse Autoencoder Auto-Interpretability Mar 2026 – Present

- Built an LLM auto-interpretability pipeline (Claude Haiku explainer, GPT-4o-mini scorer) for Gemma-2-2B sparse autoencoder (SAE) features and ran it **10,000 times**, finding that no feature kept a single label across reruns.
- Traced instability to explainer example selection via a $2 \times 2 \times 2$ factorial design; fixing top- k contexts cut it by **47%** (95% CI 43–51%).

Deep Climate Learning Mar 2025 – Jun 2025

- Built CNN-MLP and ClimaX models in PyTorch Lightning; ranked **2nd of 83 teams**, cutting RMSE **46%** vs. baseline.